



Article

Agentic AI Framework to Automate Traditional Farming for Smart Agriculture

Muhammad Murad ¹, Muhammad Ahmed ¹, Nizam ul din ¹, Muhammad Farrukh Shahid ¹,
Shahbaz Siddiqui ¹, Daniel Byers ², Muhammad Hassan Tanveer ^{2,*} and Razvan C. Voicu ²

¹ FAST School of Computing, National University of Computer and Emerging Sciences (FAST-NUCES), Karachi 75400, Pakistan; pitafimurad99@gmail.com (M.M.); muhammadahmed12072@gmail.com (M.A.); nizamudinqureshi@gmail.com (N.u.d.); shahbaz.siddiqui@nu.edu.pk (S.S.)

² Department of Robotics and Mechatronics Engineering, Kennesaw State University, Marietta, GA 30144, USA; dbyers10@students.kennesaw.edu (D.B.)

* Correspondence: mtanveer@kennesaw.edu

Abstract

Artificial intelligence (AI) shows great promise for transforming the agriculture sector and can enable the development of many modern farming practices over conventional methods. Nowadays, AI agents and agentic AI have attained popularity due to their autonomous structure and working mechanism. This research work proposes an agentic AI framework that integrates multiple agents developed for farming land to promote climate-smart agriculture and support United Nations (UN) sustainable development goals (SDGs). The developed structure has four agents: Agent A for monitoring soil properties, Agent B for weather sensing, Agent C for disease detection vision sensing in rice crops, and Agent D, a multi-agent supervisor agent chatbot connected with the other agents. The overall objective was to connect all agents on a single platform to obtain sensor data and perform a predictive analysis. This will help farmers and landowners obtain information about weather conditions, soil properties, and vision-based disease detection so that appropriate measures can be taken on agricultural land for rice crops. For soil properties (nitrogen, phosphorus, and potassium) from Agent A and climate data (temperature and humidity) from Agent B, we deployed the long short-term memory (LSTM), gated recurrent unit (GRU), and one-dimensional convolutional neural network (1D-CNN) predictive models, which achieved an accuracy of 93.4%, 94%, and 96% for Agent A; a 0.27 mean absolute error (MAE) for temperature; and a 2.9 MAE for humidity on the Agent B data. For Agent C, we used vision transformer (ViT), Mobile-ViT, and RiceNet (with a diffusion model layer as a feature extractor) models to detect disease. The models achieved accuracies of 95%, 98.5%, and 85.4% during training respectively. Overall, the proposed framework demonstrates how agentic AI can be used to transform conventional farming practices into a digital process, thereby supporting smart agriculture.



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Keywords: agentic AI; AI agents; artificial intelligence; agriculture; sensors; climate change

1. Introduction

Artificial intelligence (AI) has brought significant transformation to the agriculture sector, opening many advanced and innovative avenues for researchers and agricultural domain experts [1]. The growing use of AI in agriculture benefits not only farmers, but the entire food chain. Undoubtedly, the integration of cutting-edge technologies into

conventional farming practices is essential to meet the demands of rapid population growth, with the population expected to reach 10 billion by 2050 [2]. Rice is one of the major crops of Pakistan, contributing 8% to global rice production and 10% to the world's rice exports [3]. However, rice cultivation emits CH_4 (methane), N_2O (nitrous oxide), and other greenhouse gases (GHGs), which significantly contribute to global warming [4]. As outlined by the International Rice Research Institute (IRRI) [5], proper land preparation is an essential step for rice cultivation; the land must be adequately leveled, and the soil moisture and conditions must be carefully monitored. AI has notably transformed various farming processes into digitized and automated frameworks, including soil tillage and seeding, weed detection and monitoring, crop yield prediction, harvest monitoring, livestock management, irrigation control, and more [6]. The findings of this study align with UN SDGs 2 (Zero Hunger), 8 (Decent Work and Economic Growth), 13 (Climate Action), and 15 (Life on Land) [7]. Robotic machines and autonomous systems are increasingly being used in agriculture to automate production processes and overall farm management. This adoption accelerates the crop production cycle and minimizes unnecessary delays [8].

Recently, AI agents and agentic AI have gained considerable attention in fields such as healthcare, transportation, finance, smart cities, agriculture, and more [9]. Today, AI agents are capable of autonomous learning to enhance their performance and can intelligently interact within dynamic environments [10]. The distinction between AI agents and agentic AI lies in the fact that traditional AI agents function as single-entity systems designed to achieve goal-oriented tasks, while agentic AI consists of multiple agents that communicate and coordinate with one another to perform interrelated subtasks dynamically [11–13]. Although AI technologies have been widely explored in agriculture, most existing studies have focused on isolated functionalities such as soil analyses [14], climate predictions [15], or vision-based disease detection [16]. These approaches lack a unified framework that enables autonomous, multi-dimensional sensing and coordinated decision-making across field environments. Moreover, the current agricultural robotics research often emphasizes hardware development without integrating intelligent multi-agent coordination or real-time cross-modal data fusion [17]. In the context of rice cultivation, particularly in developing countries like Pakistan, there remains a notable gap in systems capable of simultaneously monitoring soil nutrients, climate conditions, and plant health while providing actionable insights through natural language interfaces. The existing studies on agent-based agricultural systems have rarely incorporated modern large language models (LLMs) to facilitate autonomous reasoning, farmer interaction, or seamless integration with heterogeneous field agents [18,19]. To address these limitations, this study proposes a comprehensive agentic AI framework that integrates soil, climate, and vision-sensing agents into a coordinated multi-agent architecture supported by a locally developed agricultural robot. Furthermore, we introduced an LLM-powered chatbot built on the Gemini framework [20], enabling the real-time interpretation of sensor data, natural language interaction, and autonomous decision support. This integrated approach fills the identified research gap by providing a practical, field-ready, multi-agent system capable of supporting modern precision agriculture and improving the rice farming efficiency [21]. Figure 1 shows a conceptual view of the proposed work. Nevertheless, the main contributions of this work are mentioned below:

1. The proposal of an agent-based framework for smart agriculture.
2. The implementation of a multi-agent chatbot using an LLM for various tasks such as climate sensing, soil property monitoring, and vision-based rice plant disease detection.
3. The establishment of communication between Agent A (soil sensing), Agent B (weather sensing), and Agent C (vision sensing), all connected to a supervisor agent.

4. The development of a complete dashboard to receive and visualize sensor data.
5. An analysis and evaluation using loss and accuracy plots.

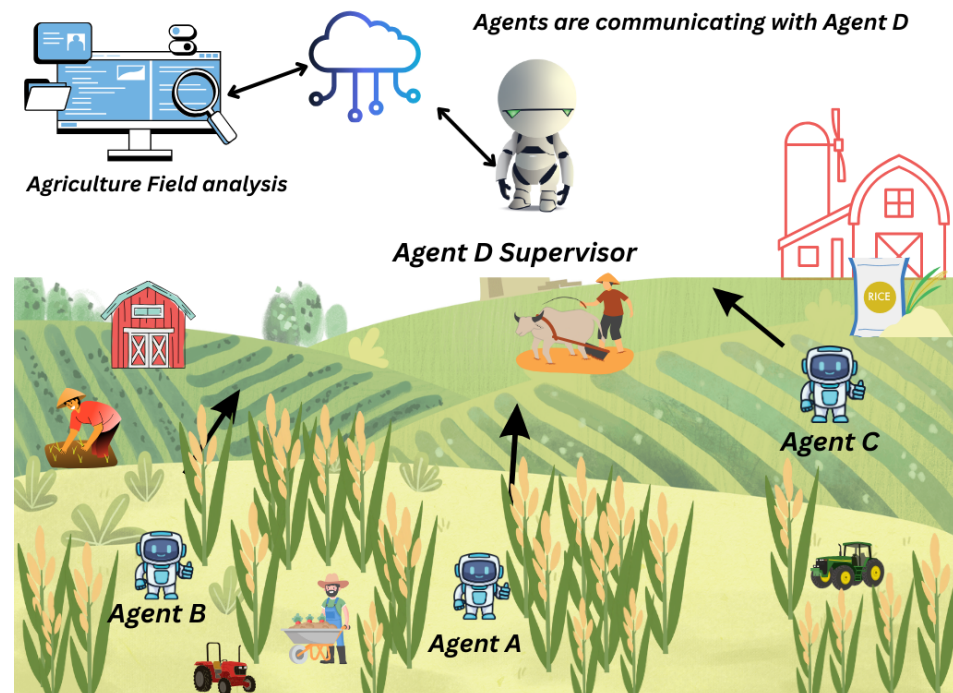


Figure 1. Conceptual figure illustrating a coordinated agricultural multi-agent system deployed across a crop field. Field agents (A, B, and C) perform localized sensing and operational tasks while transmitting data to Agent D, the supervisory controller. Agent D interfaces with a cloud-enabled agriculture field analysis platform to support decision-making and farm management. The diagram emphasizes hierarchical communication, field coverage, and agent collaboration.

1.1. Problem Statement

Modern agriculture is rapidly transitioning toward data-driven and autonomous farming systems due to increasing climate uncertainty, soil degradation, labor shortages, and the growing demand for sustainable crop production. Although recent studies have highlighted the role of AI in soil assessments [14], climate predictions [15], and plant disease detection [16], the existing solutions remain siloed and unable to operate collectively in real-time field environments. Furthermore, the current agricultural robotics and AI approaches lack a unified multi-agent or agentic AI framework capable of integrating heterogeneous sensing modalities, coordinating tasks autonomously, and supporting scalable farm automation [17,18]. These limitations led to the following research questions that guided this study:

1. How can AI be used to make data-driven decisions in farming using soil health, crop conditions, and weather patterns? Prior studies have emphasized the need for integrating soil, climate, and crop indicators into unified decision support systems [14,15].
2. How can agentic AI automate communication and workflow among multiple agricultural agents? Recent surveys have indicated that multi-agent systems in agriculture remain limited in autonomy, coordination, and real-time communication [18].
3. What computer vision models can efficiently detect diseases in rice plants for real-time field deployment? Deep learning advancements exist, but few models are optimized for practical rice-field scenarios [16].
4. How can weather forecasting be leveraged to support early decision-making and prevent agricultural losses? Studies have highlighted challenges in integrating predictive climate analytics directly into automated farm decisions [15].

Therefore, there is a compelling need to develop an agentic AI-based agricultural framework that unifies soil sensing, climate forecasting, and vision-based disease detection into a coordinated multi-agent system capable of autonomous communication, optimized workflow distribution, and real-time decision support for smart rice farming.

1.2. Major Contributions

This study makes notable contributions to address the environmental impact of the agriculture industry with the integration of AI for a sustainable environment:

- **Agentic AI framework integration:** A multi-agent AI framework is proposed in this study that integrates soil, weather, and vision agents to enable context-aware decision-making.
- **Comparative analysis of sequential models on soil data:** Multiple soil analysis models, such as the LSTM, 1D-CNN, and GRU models, were trained to evaluate their performance on real-time sensor data and identify which model performs best.
- **Environmental awareness through sensor data:** This work analyzed how temperature and humidity data alone can be useful in training weather forecasting models that support real-time environmental awareness.
- **Integration of vision and agentic AI:** The performance of ViT, mobileViT, and diffusion models, combined with RiceNet, was evaluated on a rice disease dataset and integrated into a voting classifier within the agentic AI framework.
- **Real-time guidance for farmers:** Real-time guidance is provided to farmers through an interactive chatbot and agronomists in regional languages using state-of-the-art large language models and generative AI techniques.

This study bridges the gap between traditional decision-making in agriculture and AI-enabled environmental intelligence with the integration of a multi-agent framework for soil, weather, and vision. It further evaluated the performance of different deep learning algorithms on crop and soil datasets while providing real-time assistance to farmers using a multilingual chatbot and agronomists' advice to provide a sustainable agriculture environment. The rest of this paper is organized as follows: Section 2 presents a literature review aligned with our work; in Section 3, the system overview is covered; in Section 4, the workflow of the multi-agent framework are presented; in Section 5, the experiment and results are covered; in Section 6, discussion and future work is covered; and in Section 7 we conclude our work.

2. Literature Review

The world's population is projected to reach 10 billion by 2050, creating an urgent need for effective and sustainable farming solutions [22]. Traditional agricultural practices are becoming insufficient, making intelligent systems essential for optimizing food production and resource utilization [8]. As noted by Rosana Cavalcante de Oliveira et al. [22], modern agriculture faces increasing challenges such as climate change and supply chain disruptions. In response, researchers like Nazish Aijaz et al. [1] highlight the role of AI-driven precision farming, where real-time data from Internet of Things (IoT) devices enhances crop yield and reduces waste. Similarly, Mohammed A. Hamed et al. [23] discussed the use of AI-based predictive analytics to forecast yields and detect plant diseases at early stages, thereby improving decision-making and minimizing risks. In parallel, Din et al. [24] addressed the challenge of rice-grain adulteration by introducing RiceNet, a deep CNN model that surpasses traditional techniques, achieving up to a 94% accuracy in classifying five rice grain varieties. Their research demonstrates how deep learning-based solutions can improve traditional farming and labor-intensive agriculture problems. Sumaira Ghazal et al. [25] gave an overall overview of computer vision for agriculture, indi-

cating its application throughout the digital life cycle, from image acquisition to treatment planning, but they also identified the difficulty of integrating these models in real-world settings due to environmental instability and the lack of available facilities, which restrict complete automation. Cinar et al. [26] reported that the support vector machine (SVM) achieved an accuracy of 81.5% and a precision of 83.4%, indicating a moderate performance. Among the deep learning models, GoogleNet demonstrated superior results, with an accuracy of 89.4% and a precision of 87.3%. On the other hand, however, their proposed model included deep AlexNet achieving an accuracy of 96.1% with a precision of 94.3%, reflecting better modeling in comparison to GoogleNet. Building on these advancements, Kuska et al. [27] emphasized the revolutionizing potential of LLMs for agriculture. They contended that LLMs are better, domain-specific search engines that return expert-level answers. This is consistent with Arcila's [28] perspective on ChatGPT (GPT-4) (Chat-Generative Pre-Trained Transformer) as a fusion of search engine and smart assistant that can provide targeted and context-aware farming guidance. For example, LLMs can augment farmer consultations by providing accurate guidance on crop growing, equipment, breeding, or plant pathology directly to users, even without the presence of human experts. This is most relevant in the case of smart agriculture systems, where autonomous decision-making relies on real-time sensor input and deep domain knowledge. To clearly highlight how existing studies align with the core challenges addressed in this research, the reviewed literature was systematically mapped to the four research questions, as summarized in Table 1, demonstrating how prior work motivates the need for an integrated agentic AI framework for smart farming.

Table 1. Mapping of literature to research questions in agentic AI-based smart farming.

Research Question	Key Findings from Literature	Reference
RQ1: Data-driven decision-making using soil, crop, and weather data	AI and IoT enable real-time sensing for improved yield, reduced waste, and predictive analytics for disease and yield forecasting. Climate change and supply disruptions increase the need for data-driven agricultural decisions.	Aijaz et al. [1]; Hamed et al. [23]; Oliveira et al. [22]; Talaviya et al. [8]
RQ2: Agentic AI for multi-agent communication and workflow automation	Existing agricultural AI systems lack autonomous coordination. Multi-agent frameworks with LLMs improve decision-making, enable farmer consultation, and support workflow delegation.	Kuska et al. [27]; Arcila [28]; Patel and Oliveira [18]
RQ3: Vision models for rice disease detection	Deep learning models such as CNNs, GoogleNet, and AlexNet outperform traditional ML methods for rice disease and grain classification. Deployment remains challenging in unstable real-world environments.	Din et al. [24]; Ghazal and Munir [25]; Cinar and Koklu [26]
RQ4: Role of weather forecasting in proactive decision-making	AI-driven climate and weather forecasting reduces risk and improves planning, supporting proactive decision-making and early interventions.	Fernandez and Gupta [15]; Oliveira et al. [22]

After identifying the research questions and the corresponding gaps highlighted in the literature, it became clear that the existing agricultural AI systems remain fragmented, lacking a unified agentic framework capable of autonomous sensing, decision-making, and field-level action. To address these limitations, we propose an integrated agentic AI system that extends precision agriculture toward full autonomy. At the core of this system is our

custom-designed Mali Robot, developed specifically to automate critical farming operations based on real-time sensor feedback and logic derived from a large language model (LLM). The Mali Robot autonomously irrigates crops when the soil moisture levels are low, applies pesticides when rice leaf disease is detected, and issues alerts during unusual weather conditions. A multi-agent chatbot also gives farmers real-time, language-adaptive advice based on live data from soil, climate, and vision agents. This system, which uses sensing, thinking, and acting, shows how smart AI can change regular farming into a completely automated, data-focused environment.

Research Objectives

Sustainability in agriculture requires innovation and an improved approach to replace traditional farming practices that harm the environment. AI is capable of achieving this by optimizing current farming practices and introducing efficient techniques that promote proper resource management, a sustainable environment, and less human effort. Thus, the objectives of this study were as follows:

- Develop an AI system that conducts real-time soil analyses using sensor data to estimate the soil health by measuring the moisture levels and nutrient content in the soil.
- Integrate computer vision models to detect and classify diseases in rice leaves, supporting farmers in the application of treatment at an early stage.
- Design an agentic AI framework that facilitates communication and workflow among soil, weather, and vision agents to achieve context-aware decision-making.
- Evaluate and improve AI models to ensure that they provide accurate results and are reliable under different environmental situations.

From these objectives, this study proposes the following scope and rationale: Use AI and deep learning models to help the agricultural industry improve its productivity, minimize and optimize resource usage, boost the economy, and reduce the impact on the environment, thus contributing to the growth of sustainable agricultural practices and smart farming.

3. System Overview

The suggested system presents an agentic AI-based smart farming system that includes monitoring, analyzing, and responding to real-time agricultural conditions automatically. The system consists of three main elements: (1) sensing agents that are distributed on the farmland, (2) a supervisory agent that performs reasoning and decision coordination, and (3) an intelligent agricultural robot, the Mali Robot, which was developed as a physical representation of autonomous field action. All of these elements form a closed-loop ecosystem in which the flow of data, decision-making, and physical actions take place in mutual relation and continuously. The sensing layer is composed of three expert agents, namely Agent A (soil sensing), Agent B (weather sensing), and Agent C (vision-based crop monitoring). The individual agents gather real-time sensor data on low-cost, but reliable, sensors and send the information to the central supervisory agent. The supervisory agent takes in diverse data streams, makes inferences with the help of a large language model (LLM), and issues advisory or operational instructions. The core of the execution layer is the so-called Mali Robot, which is a mobile platform that is designed specifically to handle autonomous operations in rice fields. The robot was constructed using local hardware, with the addition of soil moisture sensors, a pesticide-spraying unit, a water pump integration unit, a motorized chassis, and an embedded microcontroller to support local decision-making. The supervisory agent controls the robot, and it is physically autonomous, as it irrigates crops, identifies and controls rice diseases, and alerts when the environment is unfavorable.

The system is based on a multi-agent chatbot, which is driven by the latest state-of-the-art LLM technology. It allows farmers to obtain recommendations tailored to them, real-time notifications, and actionable insights in local languages. The chatbot interacts directly with all the sensory agents as well as the Mali Robot, thus facilitating transparent and user-friendly decision-making.

3.1. Proposed Framework

This section contains the dataset overview and multi-agent framework for each agent. Figure 2 shows the proposed methodology. The workflow steps in Figure 2 are defined below:

1. Soil and weather inputs are sensed, including a rice crop plant image.
2. Data input from Step 1 are uploaded to the cloud.
3. Data from the cloud in Step 2 are sent to user smart devices.
4. From Step 3, the data are sent to the supervisor agent to handle user queries.
5. From Step 4, the supervisor agent's response is sent back to user.
6. From Step 5, the response is stored in a database containing agent behaviors.

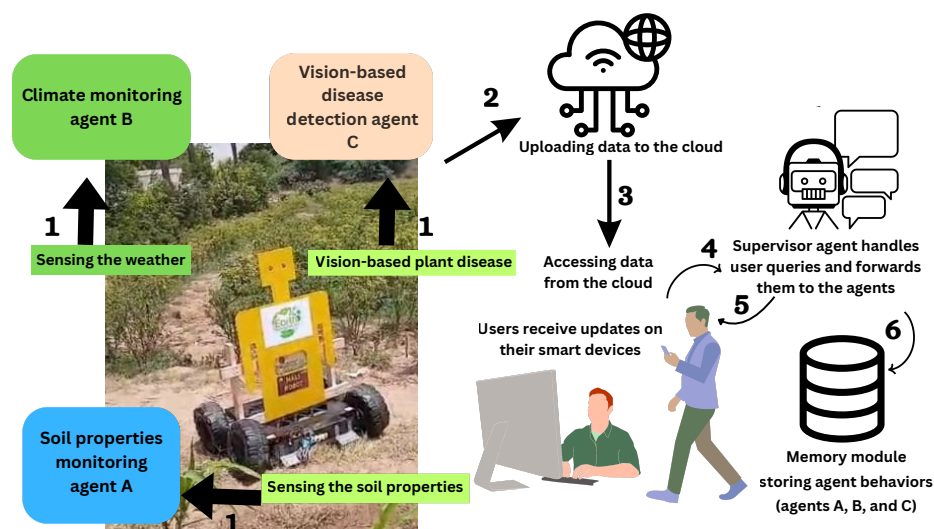


Figure 2. Proposed methodology of the agentic AI framework for smart agriculture.

The proposed agentic AI system was aimed at changing traditional farming processes into coordinated, autonomous, and data-driven ones. The architecture shown in Figure 2 has four critical layer: sensing, decision-making, action, and interaction that combine to make up a system of unified agentic AI where sensing, reasoning, and acting work together to establish fully autonomous farming. The efficiency of any of the elements is, however, strongly dependent on the quality and the variety of data utilized throughout the system. Specifically, proper soil measurements, weather forecasts, and crop imagery are required to instruct the supervisory agent, allow the Mali Robot to complete specific field tasks, and ensure that the multi-agent chatbot offers context-sensitive advice. Thus, the datasets used in the process of operationalizing the proposed framework must be described to perform soil sensing, climatic analyses, and a method of detecting rice diseases with the help of vision. The section below gives a more in-depth account of the datasets that were used in this research, as well as data acquisition, pre-processing, and integration within the agentic AI pipeline.

3.2. Mali Robot Manufacturing Overview

The Mali Robot was designed as a low-cost, modular agricultural platform capable of operating in rural farming environments. The manufacturing process consists of three major stages: mechanical fabrication, electrical integration, and system-level assembly.

The robot's frame was constructed using a combination of lightweight wooden and metal components to ensure structural stability while keeping the manufacturing cost low. Based on the design shown in Figure 3, the chassis included the following:

- A vertical humanoid-style frame for mounting sensors at different heights.
- A reinforced base plate to house onboard electronics and batteries.
- High-traction, large-diameter wheels to support movement on uneven agricultural terrain.

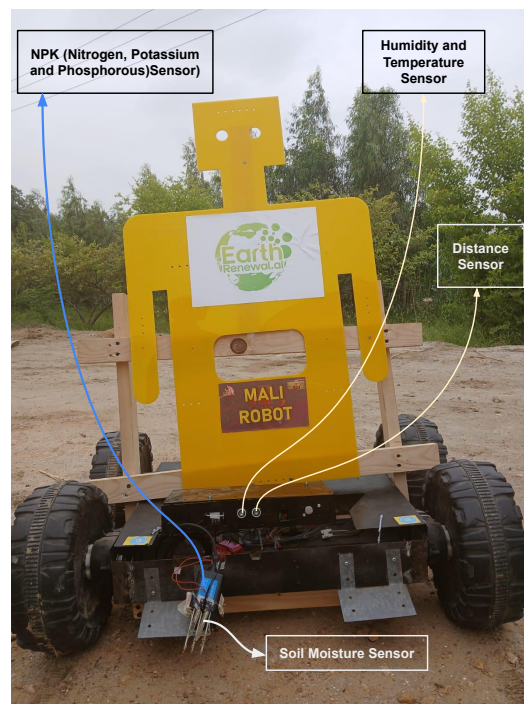


Figure 3. Sensors on Mali Robot for soil and weather agents, showing NPK, soil moisture, humidity, temperature, and distance sensors.

All frame components were manually cut, drilled, and assembled using bolts and metal brackets. The modular layout allows for the easy replacement of sensors and mechanical upgrades. The robot integrates multiple sensing modules connected to a microcontroller platform (e.g., Arduino or ESP32). Each sensor was independently wired to the control board through shielded cables. A rechargeable lithium battery powered the entire system, including the motors, sensors, and communication modules. Cable routing was performed beneath the robot's body to prevent physical obstruction during field operation.

During the final assembly stage, all sensing modules were mounted in their optimal positions on the robot frame, with the NPK and soil moisture sensors placed near the lower section, the distance sensor positioned at the front for obstacle detection, and the humidity–temperature module installed on the upper portion of the robot. An onboard enclosure was added to securely house the microcontroller and wireless communication units, ensuring protection from environmental exposure during field operation. After hardware installation, calibration procedures were performed for the NPK sensor, soil moisture probe, and environmental sensing modules to ensure accurate data acquisition before deployment. This modular manufacturing approach enhances the Mali Robot's afford-

ability, ease of maintenance, and adaptability, enabling it to perform multiple agricultural sensing tasks efficiently. Table 2 presents a detailed overview of the sensors incorporated into the Mali Robot, including their measurement ranges and operational significance, thereby illustrating how each component contributes to the robot's agricultural data acquisition framework.

Table 2. Detailed specifications of the sensors used in the Mali Robot [29].

Sensor	Specifications/Measured Values	Meaning and Purpose
NPK Sensor (Nitrogen, Phosphorus, Potassium)	- N: 0–1999 mg/kg - P: 0–1999 mg/kg - K: 0–1999 mg/kg - Output: Analog/Digital	Measures the soil content of nutrients essential for rice growth. Higher nitrogen improves leaf development, phosphorus strengthens roots, and potassium enhances plant immunity.
Soil Moisture Sensor	- Moisture Range: 0–100% - Output Voltage: 0–3.3/5 V - Probe Length: 60–100 mm	Determines the water percentage present in soil. Values < 20% indicate dry soil, while values > 70% indicate water saturation. Used for irrigation control and water management.
Humidity and Temperature Sensor (DHT22/AM2302)	- Temperature: −40 to 80 °C - Humidity: 0–100% - Accuracy: ±0.5 °C, ±2%	Monitors environmental conditions to interpret microclimate behavior, disease probability, and evapotranspiration rate.
Distance Sensor (Ultrasonic HC-SR04)	- Range: 2–400 cm - Accuracy: ±3 mm - Frequency: 40 kHz	Measures the distance to obstacles in front of the robot. Used for navigation, obstacle avoidance, and safe movement across fields.

3.3. Real-Time Data Acquisition for Predictions

The Mali Robot constantly gathers information in real time that is related to the environment and soil conditions via the sensing modules located throughout the body of the robot. All the sensors transmit their live data to the onboard microcontroller, which functions as the processor for the data acquisition. The NPK sensor and soil moisture probe, placed at the bottom of the frame, are used to detect the nutrient concentration and the water content in the soil at the surface of the field. At the same time, atmospheric data are measured by the temperature and humidity module on the upper part, and the distance sensor on the front detects obstacles and the spatial layout during navigation. All of the obtained signals are processed and digitized via the analog-to-digital conversion pipeline of the microcontroller. The information is subsequently sent in real time via a wireless communication interface through WiFi, and can be easily incorporated into the Arduino cloud-based server. The data from the cloud are transmitted to the dashboard, where they are pre-processed by normalizing, managing missing values, and filtering noise. These processed inputs are then sent to the AI-based prediction models, which consist of crop health assessments, soil condition assessments, and climate forecasting.

Dataset Overview

Two types of datasets were collected. One dataset was collected from the agriculture field of Memon Goth (Karachi), Gharo (Thatta), and Chilliya (Thatta) using the Mali Robot. The Mali Robot used sensors to sense data, through which the real-time values were uploaded onto the Arduino cloud. The cloud values were fetched to the dashboard through the Arduino API (Application Programming Interface). Figure 3 shows the sensors used on

the Mali Robot. The second dataset was the rice image dataset, which is well known and was obtained from Kaggle.

- **Soil Properties:** This dataset was collected using the Mali Robot; it collected nitrogen, phosphorus, potassium, and soil moisture data from the soil using NPK (nitrogen, phosphorus and potassium) and soil moisture sensors. The dataset contained 12,000 samples, of which 80% belonged to the training set and 20% to the validation set. Table 3 shows the data samples for the soil properties. Figure 4 shows the Mali Robot deployment in an agricultural field, including soil and weather dataset collection and a farmer survey.

Table 3. Data samples for soil and weather properties.

Timestamp	Potassium (pot)	Phosphorus (phos)	Nitrogen (nit)	Soil Moisture	Humidity	Temperature	Distance
6 August 2025 19:24	138.5	45.2	35.4	72.1	48	30	0
6 August 2025 19:23	121.2	25.9	24.7	66.9	52	32	0
6 August 2025 19:22	88.9	14.0	81.4	77.3	49	30	0
6 August 2025 19:21	108.3	40.5	40.0	47.9	40	31	0
6 August 2025 19:20	179.3	33.6	67.3	53.8	54	32	0
6 August 2025 19:19	178.2	30.0	44.7	46.1	56	30	0
6 August 2025 19:18	152.7	47.2	74.1	51.8	68	31	0
6 August 2025 19:17	88.5	11.5	45.5	64.0	63	31	0
6 August 2025 19:16	185.1	40.4	71.9	56.9	65	31	0
6 August 2025 19:15	65.9	17.5	72.6	48.0	56	32	0



Figure 4. Mali Robot deployment in an agricultural field with soil and weather dataset collection and a farmer survey in the Sindh region, Pakistan.

- **Weather Forecasting:** This dataset was collected using the Mali Robot; it collected humidity and temperature data from the environment using a humidity and temperature sensor. The dataset contained 12,000 samples, of which 90% belonged to the training set and 10% to the validation set. Table 3 shows the data samples for the weather variables. Figure 4 shows the Mali Robot deployment in an agricultural field, including soil and weather dataset collection and a farmer survey.
- **Rice Crop Images:** The rice crop image dataset comprised a well-known dataset from Kaggle (<https://www.kaggle.com/datasets/loki4514/rice-leaf-diseases-detection> (accessed on 1 November 2025)). It consisted of 10 classes (bacterial leaf blight, brown spot, healthy, leaf blast, leaf scald, narrow brown spot, neck blast, rice hispa, sheath

blight, and tungro). The image samples were split into 15,000 for the training set and 3422 for the validation set. Figures 5 and 6 show the rice image dataset samples.



Figure 5. Rice image dataset: 1 (bacterial leaf blight), 2 (brown spot), 3 (healthy), 4 (leaf blast), and 5 (leaf scald).



Figure 6. Rice image dataset: 6 (narrow brown spot), 7 (neck blast), 8 (rice hispa), 9 (sheath blight), and 10 (tungro).

4. Workflow of the Multi-Agent Framework

In this section, we describe the workflow of the proposed multi-agent framework, outlining how the soil, weather, vision, and supervisory agents collaborate to sense the environment, analyze information, and trigger appropriate farming actions. Figure 7 illustrates this workflow, showing how data flow begins at the sensing agents, is fused and interpreted by the supervisory agent, and ultimately results in autonomous decisions executed through the Mali Robot and the advisory chatbot. The following mathematical formulation further clarifies these interactions, the information flow, and the decision processes that enable the system to function autonomously under real-world agricultural conditions.

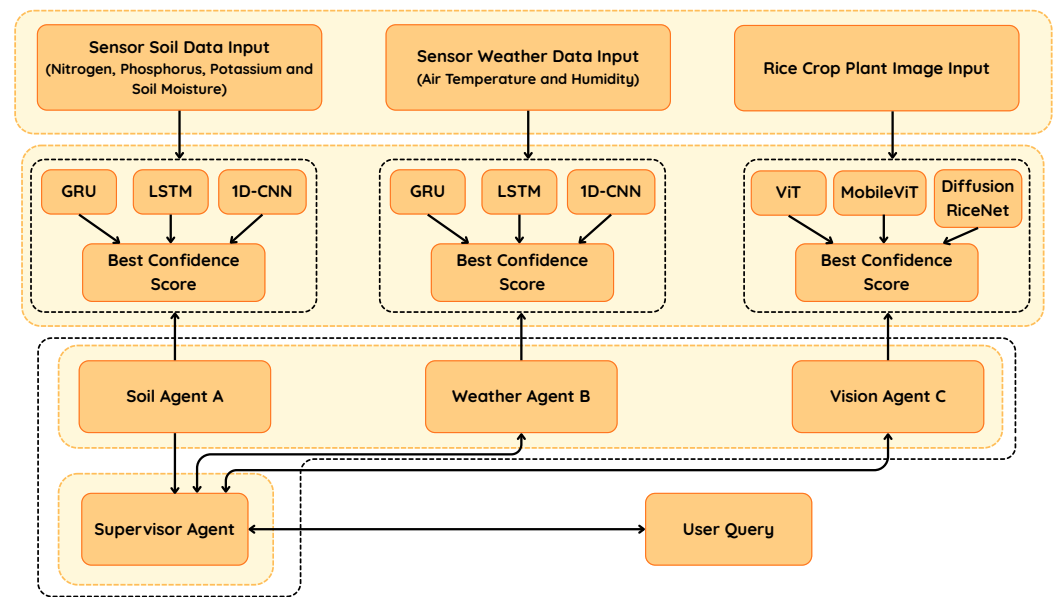


Figure 7. Workflow of the multi-agent framework.

All the experiments were conducted using Python 3.10 with deep learning frameworks, including PyTorch 2.4.1 and TensorFlow 2.18. The architectures for the GRU, LSTM, and 1D-CNN models were implemented using PyTorch, while the ViT and MobileViT models were based on the timm and mobilevit libraries. The diffusion-based model was implemented using the HuggingFace Diffusers framework (version 1.5). Training was performed on a workstation equipped with an NVIDIA RTX 4060 GPU (8 GB VRAM), an Intel Core i9-13900HX CPU, and 16 GB RAM.

4.1. Biases and Challenges in Data Collection and Labeling

Sensor-based data acquisition is subject to several potential sources of bias. Soil and weather sensors may experience calibration drift, noise from environmental fluctuations, temporary signal loss, or incomplete readings due to hardware limitations. To mitigate these issues, the data were pre-processed using smoothing filters, missing-value interpolation, and periodic sensor calibration checks during field deployment. For the vision dataset, a Kaggle dataset was used, which was already labeled into 10 different classes, leading us to conclude that there were no biases or challenges with the vision dataset.

4.1.1. Soil Agent

The soil agent uses three different models to classify the soil health conditions into a binary classification (good and poor). It identifies the prediction based on the best confidence score from each model. The models used were the GRU, LSTM, and 1D-CNN models. Later, the soil agent provides the classification to the supervisor agent, which processes the results. Algorithm 1 shows the implementation of soil agent.

GRU (Gated Recurrent Unit)

The GRU, introduced by Cho et al. [30], is a computationally efficient substitute for the LSTM that streamlines the memory control mechanisms without losing the capacity to learn long-term dependencies. The GRU combines the forget gate and input gate into an update gate, and adds a reset gate to handle information flow, which decreases the model complexity and computational expense without compromising much in performance. The GRU cell updates as per Equation (1) following the formulation introduced by Cho et al. [30]. The hidden state in the GRU is the memory unit; it allows the model to

remember previous inputs and apply that knowledge to make better predictions at each time step [30].

$$\text{GRU: } h_t = \text{GRU}(x_t, h_{t-1}) \quad (1)$$

where

$$\begin{cases} x_t & \text{is the input at time } t; \\ h_{t-1} & \text{is the hidden state at time } t-1; \\ h_t & \text{is the hidden state at time } t. \end{cases}$$

LSTM (Long Short-Term Memory)

The LSTM network, proposed by Hochreiter and Schmidhuber [31], is a particular flavor of a recurrent neural network (RNN) that has been specifically developed to solve the disadvantages of ordinary RNNs in dealing with long-range dependencies as a result of vanishing or exploding gradients. LSTM has a memory cell and three gates: the input gate, the forget gate, and the output gate that together control information flow, thus allowing the network to retain or discard information over long sequences. The LSTM network updates its hidden state as per Equation (2) based on the research conducted in [31–33].

$$\text{LSTM: } (h_t, c_t) = \text{LSTM}(x_t, h_{t-1}, c_{t-1}) \quad (2)$$

where

$$\begin{cases} x_t & \text{is the input at time } t; \\ h_{t-1} & \text{is the hidden state at time } t-1; \\ c_{t-1} & \text{is the cell state at time } t-1; \\ h_t & \text{is the hidden state at time } t; \\ c_t & \text{is the cell state at time } t. \end{cases}$$

LSTM's gating system in a well-structured way counters gradient decay over long sequences and is well suited for the analysis of sequential data tasks.

1D-CNN

The 1D-CNN is a deep learning architecture that carries out convolution on one-dimensional data, and hence, it is well suited for time-series classification. Unlike RNNs, 1D-CNNs do not require sequence processing and are capable of effectively learning local spatial patterns through the use of filters [34]. Table 4 shows the 1D-CNN architecture.

Table 4. 1D-CNN architecture overview.

Layer No.	Layer Type	Details
1	Convolutional (Conv1D)	64 filters, kernel size = 2, activation = ReLU
2	MaxPooling1D	Pool size (default or defined), reduces spatial dimensions
3	Dropout	Dropout rate (e.g., 0.5) to prevent overfitting
4	Flatten	Converts the feature map into a 1D vector
5	Dense (Hidden)	Fully connected layer, activation = ReLU
6	Dense (Output)	1 unit, activation = sigmoid (for binary classification)

Algorithm 1: Soil Algorithm: Soil health classification.

Input: Soil readings X (nitrogen, phosphorus, potassium, moisture)
Output: Soil health status $\in \{\text{good}, \text{poor}\}$

```

1  $X_{\text{scaled}} \leftarrow \text{Scale}(X)$ 
2  $p_{\text{gru}} \leftarrow \text{GRU}(X_{\text{scaled}})$ ; // Probability from GRU model
3  $p_{\text{lstm}} \leftarrow \text{LSTM}(X_{\text{scaled}})$ ; // Probability from LSTM model
4  $p_{\text{cnn}} \leftarrow \text{1D-CNN}(X_{\text{scaled}})$ ; // Probability from 1D-CNN model
5  $p_{\text{avg}} \leftarrow (p_{\text{gru}} + p_{\text{lstm}} + p_{\text{cnn}})/3$ ; // Soft voting ensemble
6 if  $p_{\text{avg}} \geq 0.5$  then
7    $\text{output} \leftarrow \text{good}$ 
8 else
9    $\text{output} \leftarrow \text{poor}$ 
10 return  $\text{output} \leftarrow \text{Supervisor Agent}(\text{SoilAgent})$ 

```

4.1.2. Weather Agent

The weather agent works by predicting the weather for the next 5 h. It uses two features: humidity and air temperature, along with timestamps from the sensors. It was trained on three different time-series models: the LSTM, GRU and 1D-CNN models. These models predict the weather for the next 5 min, and with recursive prediction, the agent predicts the weather for the next 5 h. It identifies the prediction based on the best confidence score from each model. Later, it provides the results to the supervisor agent, which processes the results. Algorithm 2 shows the implementation of weather agent.

Algorithm 2: Weather Algorithm: Environmental humidity and temperature prediction.

Input: Time-series readings X (humidity, temperature, timestamp)
Output: Predicted humidity and temperature

```

1  $X_{\text{scaled}} \leftarrow \text{MinMaxScale}(X)$ 
2  $X_{\text{seq}} \leftarrow \text{CreateTimeWindows}(X_{\text{scaled}})$ 
3  $y_{\text{lstm}} \leftarrow \text{LSTM}(X_{\text{seq}})$ ; // Prediction from LSTM model
4  $y_{\text{gru}} \leftarrow \text{GRU}(X_{\text{seq}})$ ; // Prediction from GRU model
5  $y_{\text{cnn}} \leftarrow \text{1D-CNN}(X_{\text{seq}})$ ; // Prediction from 1D-CNN model
6  $\text{output} \leftarrow (y_{\text{lstm}} + y_{\text{gru}} + y_{\text{cnn}})/3$ ; // Ensemble averaging
7 return  $\text{output} \leftarrow \text{Supervisor Agent}(\text{WeatherAgent})$ 

```

4.1.3. Vision Agent

The rice image dataset was trained on three models, ViT [35], MobileViT [36], and a diffusion model [37,38] as a feature extractor integrated with RiceNet [24]. The model works on multi-class disease detection. The agent identifies the disease based on the best confidence score from each model and provides the results to the supervisor agent, which processes the results. Algorithm 3 shows the implementation of Vision agent.

Vision Transformer (ViT)

A ViT is a deep learning model that works on the principles of transformers. These models were originally built for natural language processing (NLP) tasks to computer vision tasks; they essentially split an image into fixed-size patches (16×16), flatten them, and treat them as a sequence of tokens. The tokenized patches are passed to a transformer encoder, which learns global relationships in the entire image using self-attention mechanisms [35,39]. The key advantage of a ViT is its ability to model long-range depen-

dencies without relying on inductive biases like the locality and translation invariance inherent in CNNs. This makes ViTs more flexible and powerful, especially when trained on large-scale datasets. Table 5 shows the architecture of a vision transformer.

Table 5. Vision transformer (ViT) architecture description.

Component	Description
Input Image	224 × 224 RGB image
Patch Embedding	Splits image into 16 × 16 patches, resulting in 14 × 14 = 196 patches
Patch + Position Embedding	Each patch (3 × 16 × 16) is flattened to a 768-d vector; positional embeddings are added
CLS Token	A learnable token prepended to the patch sequence is used for classification
Transformer Encoder	12 identical blocks consisting of the following: • Multi-head self-attention (12 heads) • MLP (feedforward) with hidden dimension = 3072 • LayerNorm and residual connections
Final LayerNorm	Applied after the last transformer block
Classification Head	Linear layer mapping 768 → 10

MobileViT

MobileViT is a lightweight deep learning model that integrates the strengths of convolutional neural networks (CNNs) and ViT. Unlike standard ViTs, which directly split an image into patches and process them with self-attention, MobileViT first applies convolutions to capture local spatial features and then unfolds the resulting feature maps into non-overlapping patches. These patches are flattened and passed through a transformer encoder, enabling the model to learn global dependencies using self-attention mechanisms [36]. The key advantage of MobileViT lies in its hybrid architecture: CNNs provide strong inductive biases such as locality and translation invariance, while transformers capture long-range contextual relationships across the image. This design achieves a high accuracy while maintaining computational efficiency and a low latency, making MobileViT suitable for deployment on mobile and edge devices [38,39].

Diffusion Model (as a Feature Extractor) + RiceNet

The pre-trained variational autoencoder (VAE) from the stable diffusion v1.5 model <https://huggingface.co/stable-diffusion-v1-5/stable-diffusion-v1-5> (accessed on 30 November 2025) was utilized as a feature extractor [37]. The encoder was used to compress high-dimensional image data into semantically rich latent representations. The encoder was loaded to transform each input image size to 224 × 224 × 3 into a lower-dimensional latent vector. The vector was passed to the RiceNet classification head for multi-class classification. The RiceNet architecture was initially designed by [24], which was modified to accept the latent vectors from the diffusion model. The architecture comprises four convolutional blocks followed by two fully connected (FC) layers. Each convolutional block is structured as follows:

$$\text{Conv2D} \rightarrow \text{BatchNorm} \rightarrow \text{ReLU} \rightarrow \text{MaxPooling}$$

except for the final block, where AdaptiveAvgPool2D was used instead of max pooling to downsample the feature maps to a fixed size of 256 × 1 × 1. The output was then flattened and passed through two fully connected layers with ReLU activations and Dropout for multi-class classification.

Algorithm 3: Vision Algorithm: Rice plant disease detection.

Input: rice plant image
Output: disease classification result

```

1  $X \leftarrow \text{Rice Plant Image};$ 
2  $X_{\text{preprocess}} \leftarrow \text{preprocess}(\text{image});$ 
3  $\text{acc}_{\text{ViT}} \leftarrow \text{ViT}(\text{image});$ 
4  $\text{acc}_{\text{mobileViT}} \leftarrow \text{mobileViT}(\text{image});$ 
5  $\text{acc}_{\text{diffusion}_{\text{ricenet}}} \leftarrow \text{diffusion}_{\text{ricenet}}(\text{image});$ 
6  $\text{best}_{\text{acc}} \leftarrow \max(\text{acc}_{\text{ViT}}, \text{acc}_{\text{mobileViT}}, \text{acc}_{\text{diffusion}_{\text{ricenet}}});$ 
7  $\text{Output} \leftarrow \text{Softmax}(\text{best}_{\text{acc}});$ 
8 return  $\text{Output} \leftarrow \text{Supervisor Agent}(\text{Vision Agent});$ 
```

4.1.4. Supervisor Agent

A supervisor agent or a master agent works like a communicator between two agents; it is integrated on Gemini API. It works as follows: For instance, a user could enter a query asking the supervisor agent about the current soil properties and what action they should take. The supervisor agent will analyze to which agent (soil agent, weather agent, or vision agent) this query should be sent. It will detect that this query is for the soil agent and will send the query to the soil agent. The soil agent will accept the query and provide the results with real-time soil values and a prediction, such as “Your soil properties include nitrogen = 350, potassium = 500, and phosphorus = 450; these values show that your soil condition is good”. Algorithm 4 shows the implementation of Supervisor agent.

In order to store the agents’ behavior over a span of time, we used simple, yet effective, memory storage criteria for each agent. To begin this, we used a vector database approach in which we stored the sensing and actions of Agents A, B, and C. We set the following criteria for sensing and action processes: sensing data storage for 3 months four times a day. Accordingly, we had 368 timestamps in our database. The following equations show the timestamp and agent sensing data.

$$t_1 = \left\{ \text{agent } a_1^{(p,k,n,m)}, \text{agent } b_1^{(\text{temp,hum})}, \text{agent } c_1^{(\text{img}_n, \text{img}_u|l)} \right\} \quad (3)$$

$$t_2 = \left\{ \text{agent } a_2^{(p,k,n,m)}, \text{agent } b_2^{(\text{temp,hum})}, \text{agent } c_2^{(\text{img}_n, \text{img}_u|l)} \right\} \quad (4)$$

$$t_n = \left\{ \text{agent } a_n^{(p,k,n,m)}, \text{agent } b_n^{(\text{temp,hum})}, \text{agent } c_n^{(\text{img}_n, \text{img}_u|l)} \right\} \quad (5)$$

Algorithm 4: Supervisor Agent: Query-routing system.

Input: User query Q
Output: Response from respective agent

```

1 if  $Q$  contains keywords related to temperature, humidity, or weather then
2   | response  $\leftarrow \text{WeatherAgent}(Q)$ 
3 else
4   | if  $Q$  contains keywords related to soil, NPK, fertility, moisture then
5   | | response  $\leftarrow \text{SoilAgent}(Q)$ 
6   | else
7   | | if  $Q$  contains keywords related to plant health, disease, or images then
8   | | | response  $\leftarrow \text{VisionAgent}(Q)$ 
9   | | else
10  | | | response  $\leftarrow \text{Error: Undefined Query Type}$ 
11 return response
```

where

t_i	is the timestamp;
a_i	is the agent collecting soil nutrients and moisture;
p	is the phosphorus;
k	is the potassium;
n	is the nitrogen;
m	is the soil moisture;
b_i	is the agent measuring environmental conditions;
temp	is the temperature;
hum	is the humidity;
c_i	is the vision-based agent;
img_h	is the input image;
img_u	is the output (processed) image;
l	is the contextual condition.

5. Experiments and Results

We implemented and tested three agents, each with different functionalities: a soil agent and weather agent, which work on tabular data, and a vision agent, which works on image data. These agents are integrated and connected with the supervisor agent to create multi-agent communication.

5.1. Soil Agent

5.1.1. GRU

GRU-based architecture was used to process the soil nutrient time-series data for classification. The results of the GRU are shown in Figure 8, including the accuracy and loss graphs. The accuracy graph shows a training accuracy of 95.5% and a validation accuracy of 94%. Table 6 shows a comparison of each model.

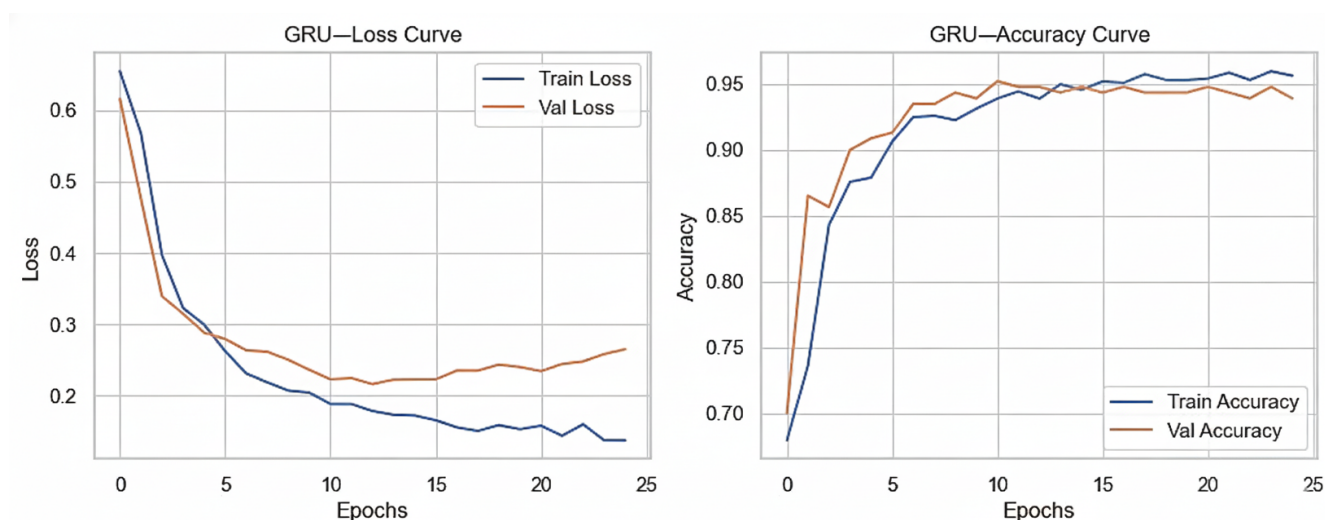


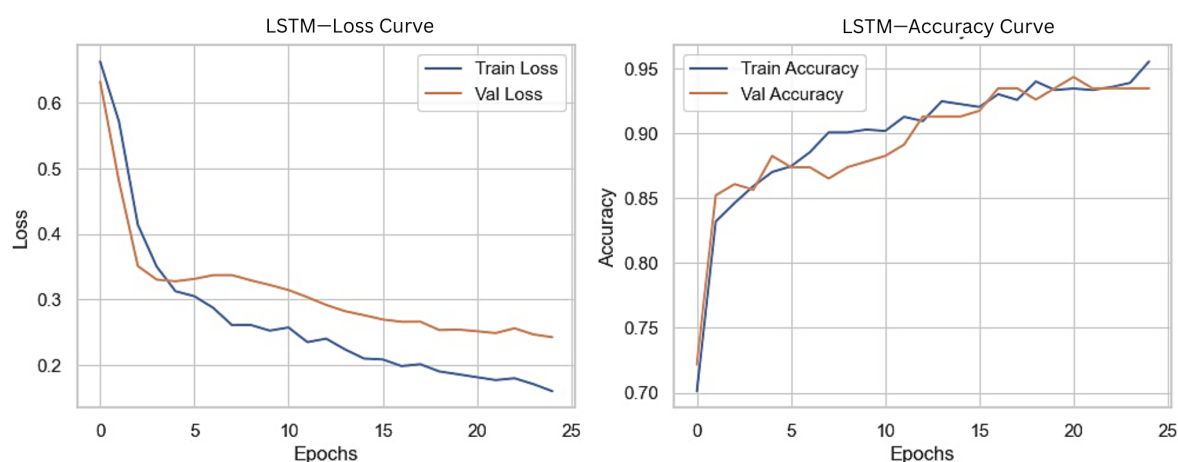
Figure 8. Soil agent: GRU accuracy and loss.

Table 6. Model comparison for soil predictions.

Model	Epochs	Training Accuracy	Validation Accuracy
GRU	25	95.8%	94%
LSTM	25	95.5%	93.4%
1D-CNN	25	97%	96%

5.1.2. LSTM

The LSTM model was applied using time-series soil nutrient inputs to learn intricate temporal dependencies and conduct a binary classification of the soil quality classes. The results of the LSTM model are shown in Figure 9, including the accuracy and loss graphs. The accuracy graph shows a training accuracy of 95.5% and a validation accuracy of 94%. Table 6 shows a comparison of each model.

**Figure 9.** Soil agent: LSTM accuracy and loss.

5.1.3. 1D-CNN

In the current research, a 1D-CNN was utilized in order to classify soil nutrient data by identifying important temporal features through the use of convolution and pooling layers. The results of the 1D-CNN are shown in Figure 10, including the accuracy and loss graphs. The accuracy graph shows a training accuracy of 97% and a validation accuracy of 96%. Meanwhile, Figure 11 shows the confusion matrix of each model. Table 6 shows a comparison of each model.

**Figure 10.** Soil agent: 1D-CNN accuracy and loss.

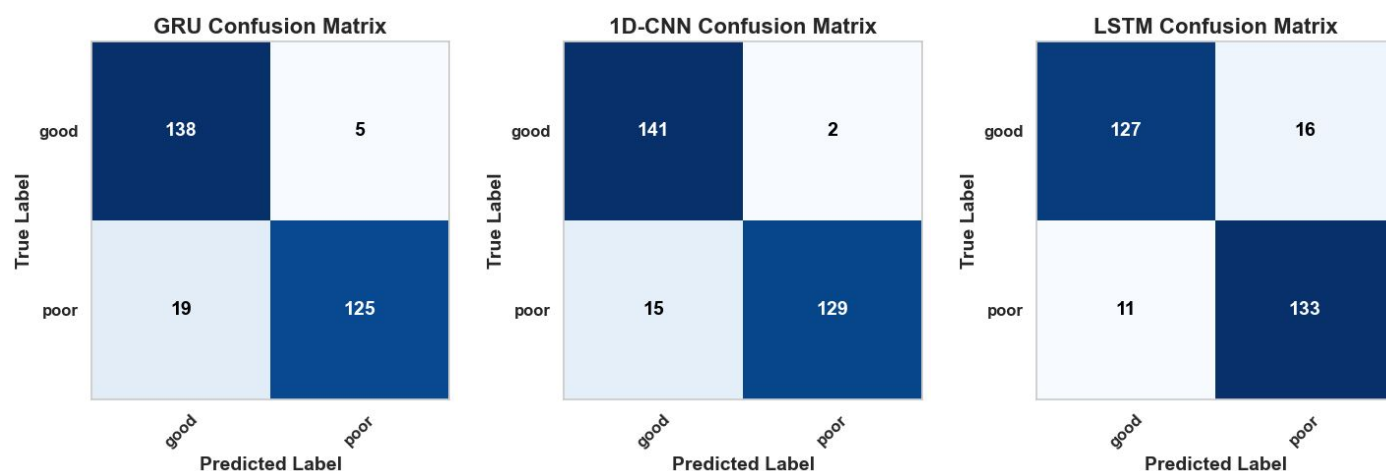


Figure 11. Soil agent: GRU confusion matrix (left), 1D-CNN confusion matrix (center), and LSTM confusion matrix (right).

5.2. Weather Agent

5.2.1. Temperature

The temperature variable was trained alongside its respective timestamps. The time-series prediction can be seen in Figure 12, which shows the actual data vs. the prediction of the applied models. To evaluate the models, the MAE score was used. LSTM reached an MAE score of 0.36, the 1D-CNN reached a score of 0.28, and the GRU had the lowest score of 0.27. The lower the MAE, the better the model was at predicting the next outcome. Figure 13 shows the MAE comparisons of each model implemented. Table 7 shows a brief comparison of three weather models.

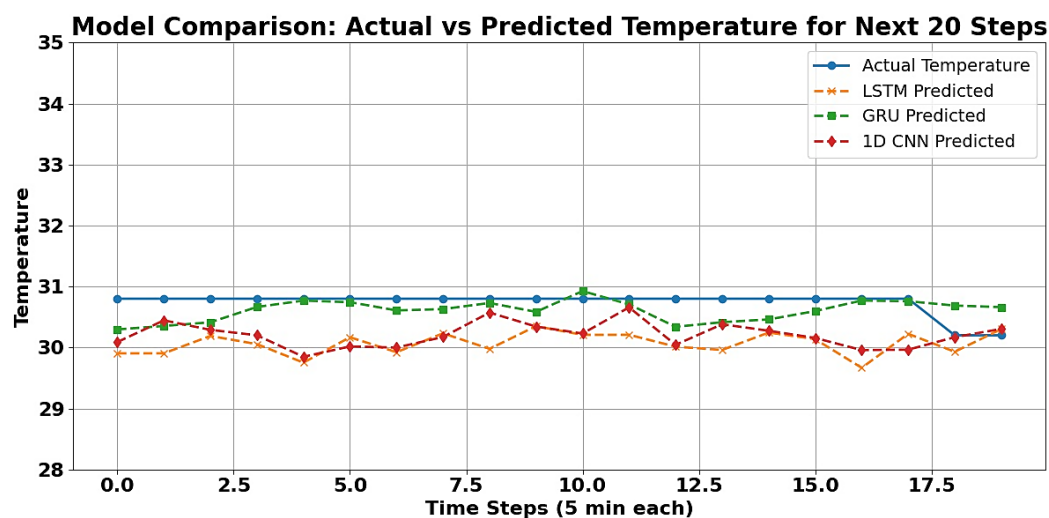


Figure 12. Weather agent: actual temperature vs. predicted temperature.

Table 7. MAE comparison of models for weather predictions.

Model	Humidity MAE	Temperature MAE
LSTM	5.7	0.36
GRU	2.9	0.27
1D-CNN	3.6	0.28

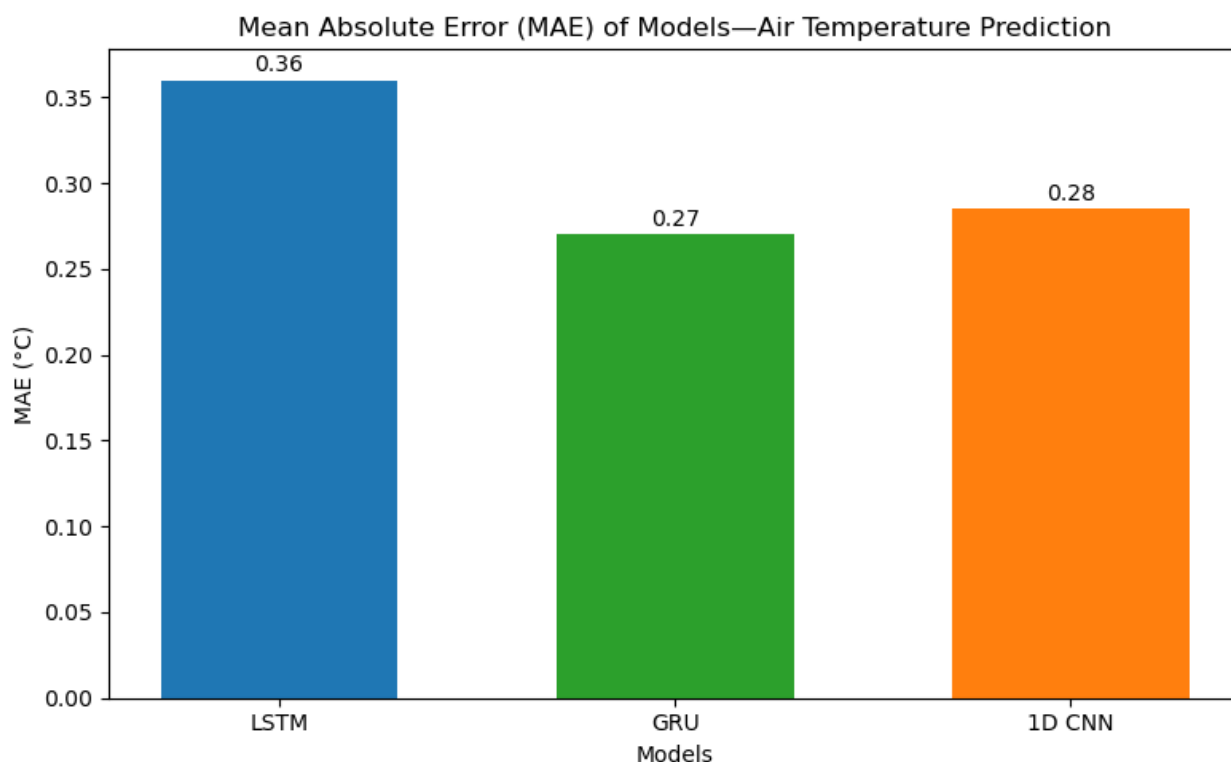


Figure 13. Weather agent: MAE comparison of temperature predictive models.

5.2.2. Humidity

The humidity variable was trained alongside its respective timestamps. The time-series prediction can be seen in Figure 14, which shows the actual data vs. the prediction of the applied models. To evaluate the models, the MAE score was used. LSTM reached an MAE score of 5.7, 1D-CNN had an MAE score of 3.6, and GRU had the lowest score of 2.9. The lower the MAE, the better the model was at predicting the next outcome. Figure 15 shows the MAE comparisons of each model implemented. Table 7 shows a brief comparison of three weather models.

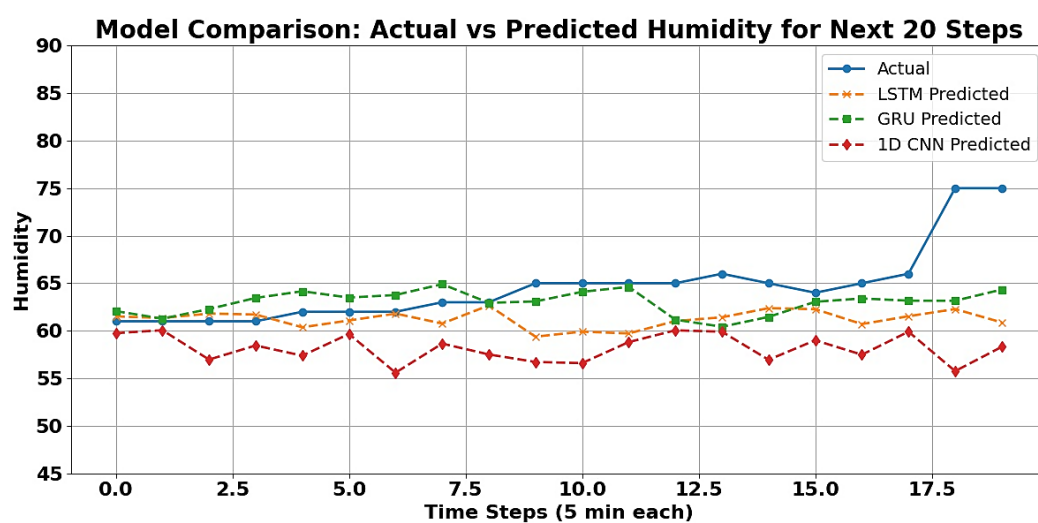


Figure 14. Weather agent: actual humidity vs. predicted humidity.

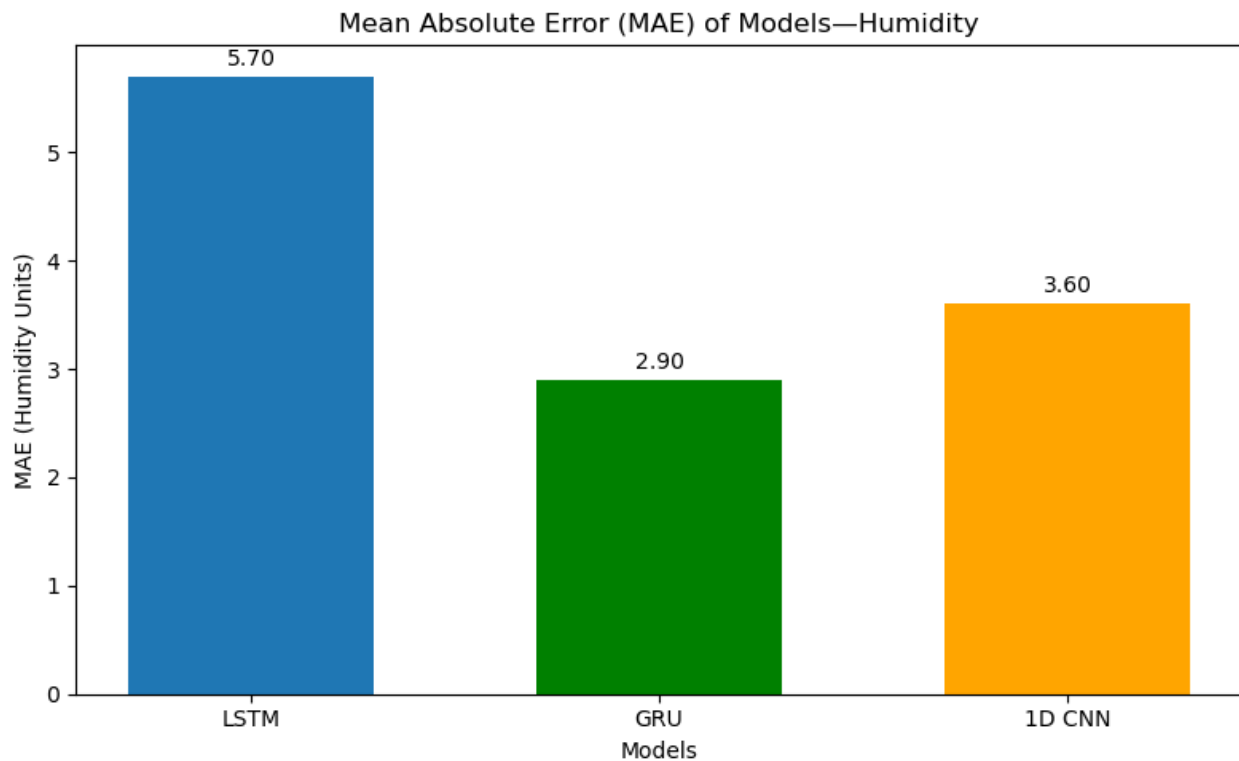


Figure 15. Weather agent: MAE comparison of humidity predictive models.

5.3. Vision Agent

5.3.1. Vision Transformer (ViT)

The ViT was trained on 10 classes of rice diseases. The transformer was trained on 10 epochs with a batch size = 32 and early stopping at patience = 4, meaning if the validation loss did not decrease for four straight epochs, then the training stopped, as this shows that the model did not improve further. Thus, the training was completed after nine epochs due to early stopping. We achieved a validation accuracy of 95%. Figure 16 shows the accuracy and loss graphs for the ViT, while Figure 17 shows the confusion matrix. We later applied cross-validation to the ViT for model stability for five k-folds and achieved an average accuracy of 92.54% with an average F1-score of 92.45%. Figure 18 shows the accuracy and F1-score of each k-fold. Table 8 shows a comparison of each deep learning model used by the vision agent.

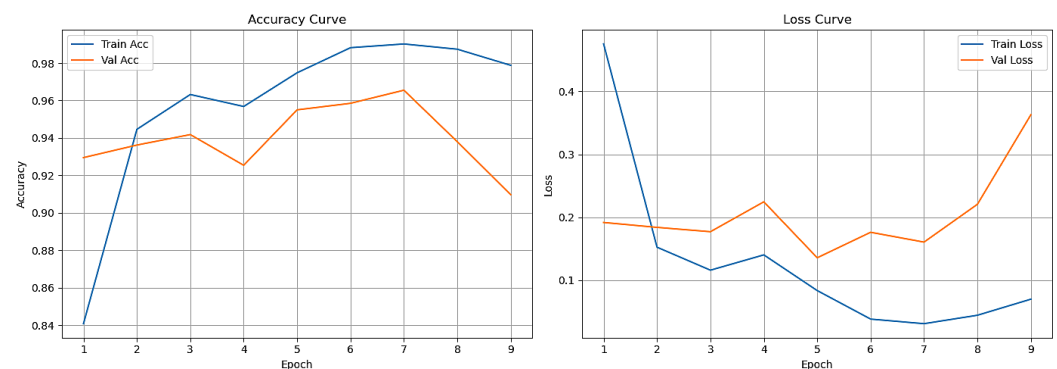


Figure 16. Vision agent: ViT accuracy and loss graph.

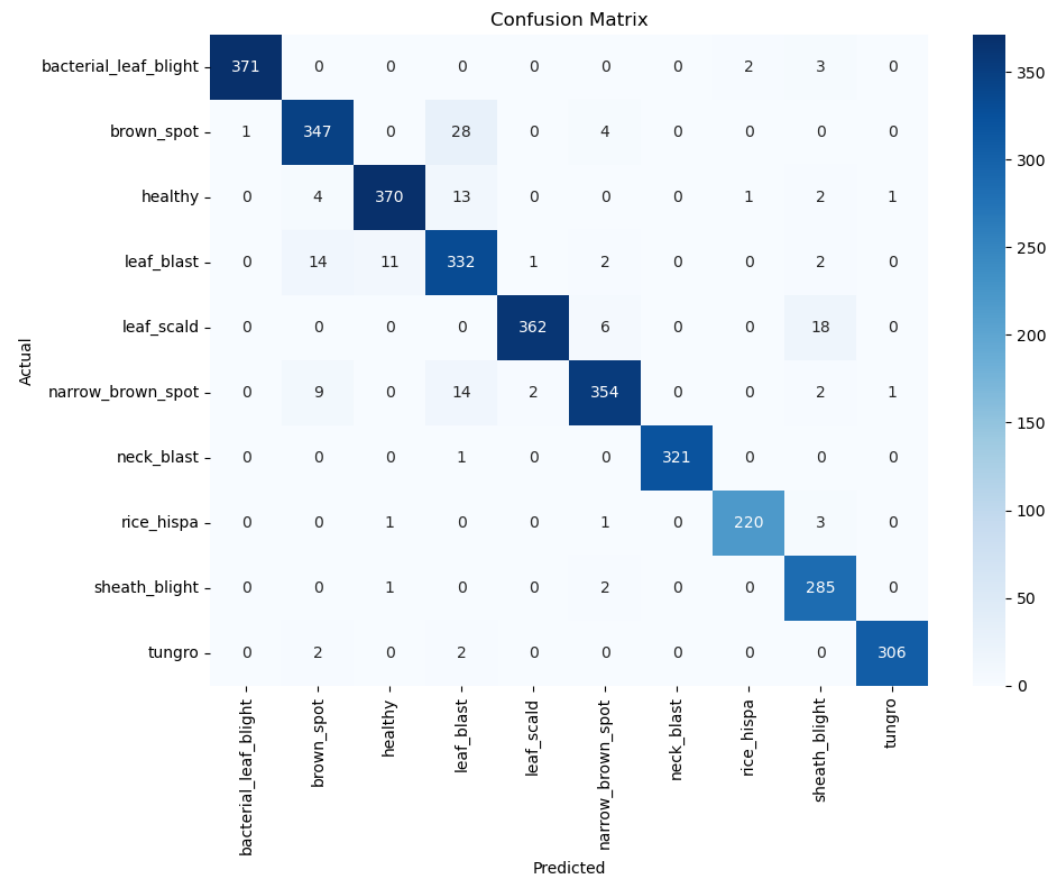


Figure 17. Vision agent: ViT confusion matrix.

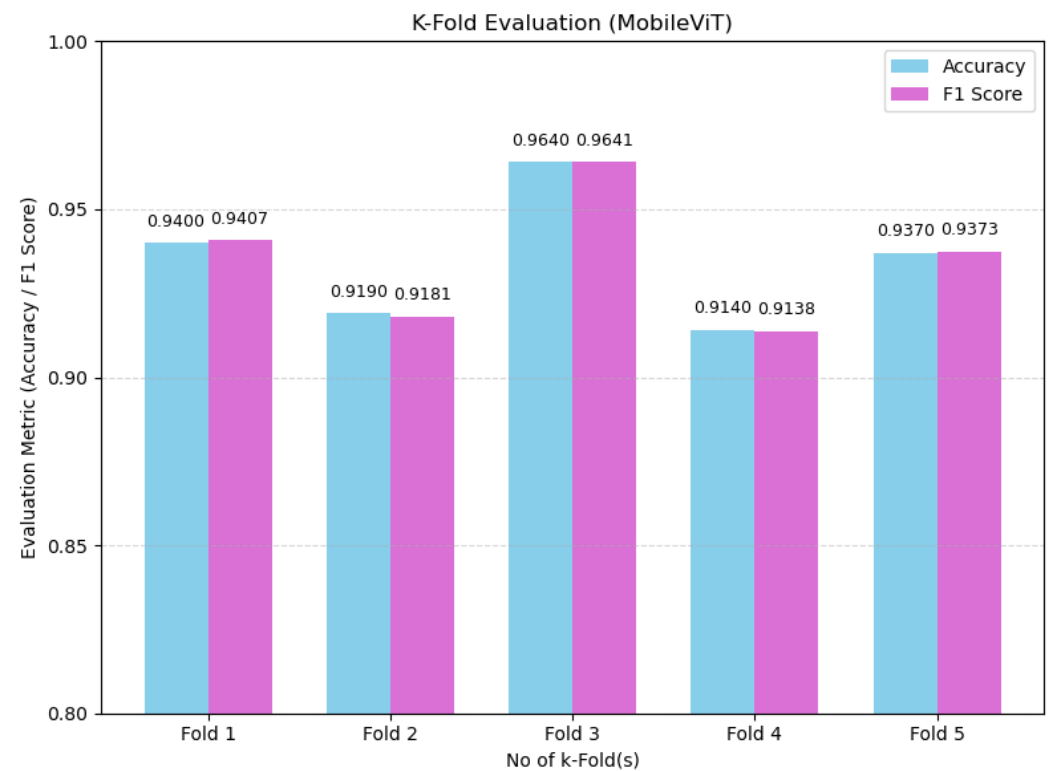


Figure 18. Vision agent: accuracy and F1-score at each k-fold.

5.3.2. MobileViT

The MobileViT was also trained on 10 classes of rice diseases. The transformer was trained with the same parameters as the simple ViT. With early stopping and patience = 4, the training was completed after 10 complete epochs. We achieved a validation accuracy of 98.51%. Figure 19 shows the accuracy and loss graphs for MobileViT, while Figure 20 shows the confusion matrix. To validate the accuracy, we applied cross-validation on MobileViT for five k-folds and achieved an average accuracy of 93.48% with an average F1-score of 93.48%. Figure 21 shows the accuracy and F1-score of each k-fold. Table 8 shows a comparison of each deep learning model used by the vision agent.

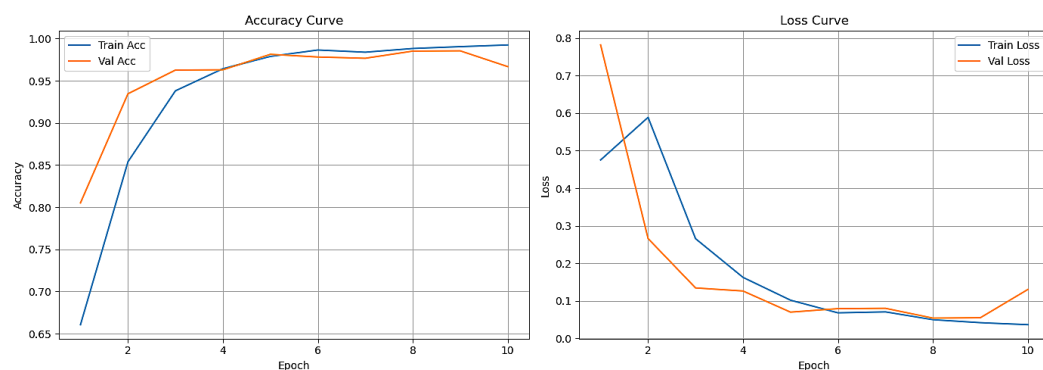


Figure 19. Vision agent: MobileViT accuracy and loss graphs.

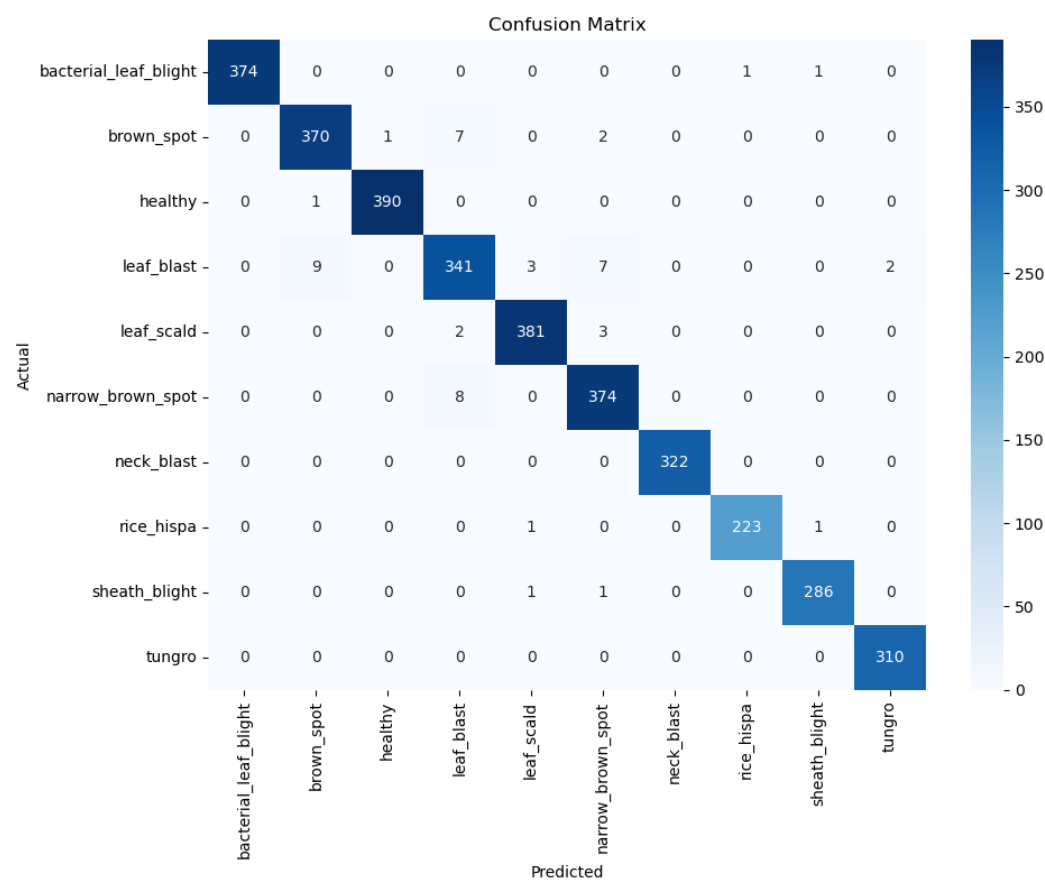


Figure 20. Vision agent: MobileViT confusion matrix.

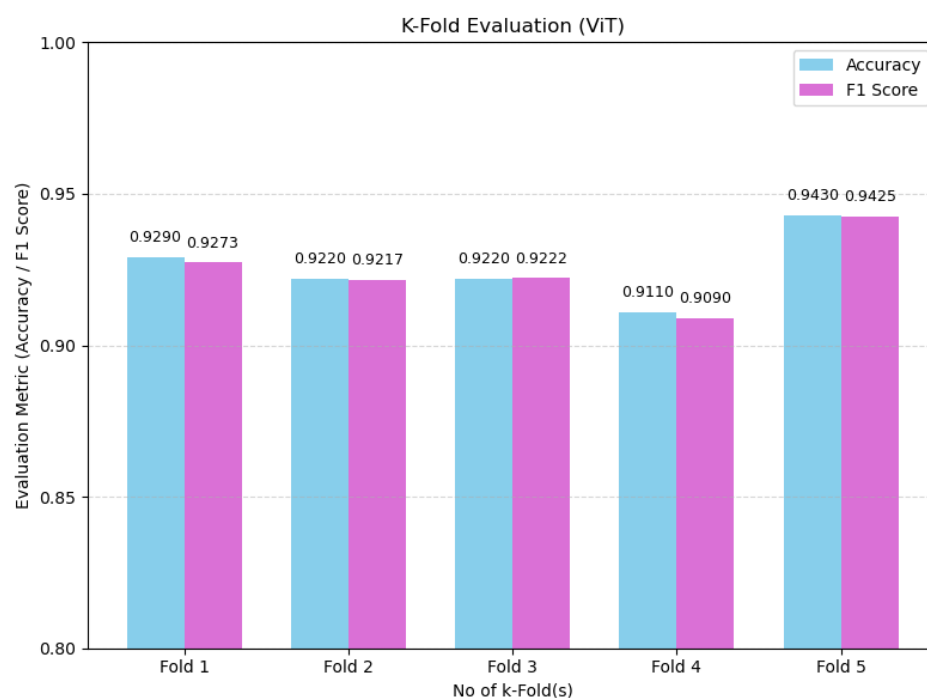


Figure 21. Vision agent: MobileViT k-folds.

5.3.3. Diffusion Model (As a Feature Extractor) + RiceNet

The diffusion model was used as a feature extractor that was fed into the RiceNet architecture. We first trained 10 epochs; however, due to a low accuracy and a potential for achieving a better accuracy at more epochs, the model was later trained on 20 epochs with the same model-fitting parameters (patience = 4). The model achieved a validation accuracy of 85.39%. Figure 22 shows the accuracy and loss graph, whereas Figure 23 shows the confusion matrix. Table 8 explains the details of the comparison of each deep learning model used by the vision agent.

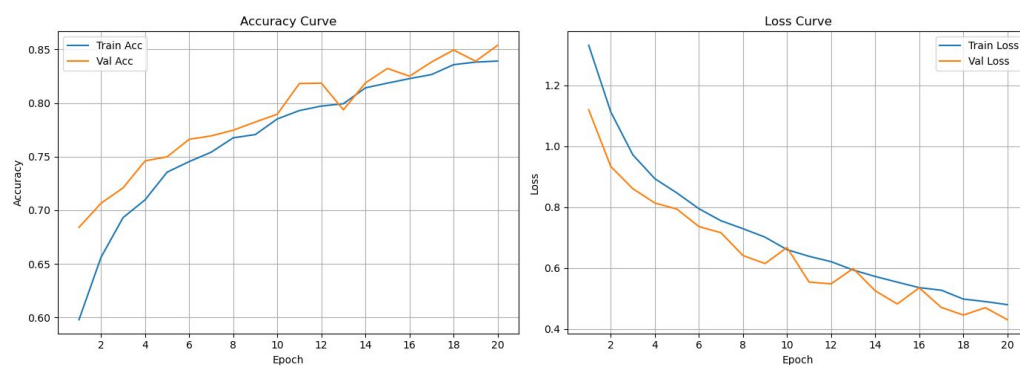


Figure 22. Vision agent: diffusion model + RiceNet accuracy and loss graphs.

Table 8. Training and cross-validation results of deep learning models.

Deep Learning Model	Epochs	Patience	Training Accuracy	K-Folds	Cross-Validation Accuracy	Cross-Validation F1-Score
ViT	10	4	95%	5	92.5%	92.5%
MobileViT	10	4	98.5%	5	93.5%	93.5%
Diffusion + RiceNet	20	4	85.4%	-	-	-

		Confusion Matrix									
Actual	bacterial_leaf_blight	358	6	0	0	3	3	0	5	1	0
	brown_spot	7	262	11	41	0	49	6	2	2	0
	healthy	1	12	331	15	0	1	1	28	2	0
	leaf_blast	1	40	38	205	39	18	1	15	1	4
	leaf_scald	0	0	0	4	360	7	0	2	13	0
	narrow_brown_spot	0	6	0	12	11	342	0	5	6	0
	neck_blast	0	0	0	0	0	0	320	1	1	0
	rice_hispa	0	1	8	1	0	6	0	205	4	0
	sheath_blight	0	0	0	23	1	18	0	5	241	0
	tungro	0	1	1	0	0	0	2	2	6	298
		bacterial_leaf_blight	brown_spot	healthy	leaf_blast	leaf_scald	narrow_brown_spot	neck_blast	rice_hispa	sheath_blight	tungro
		Predicted									

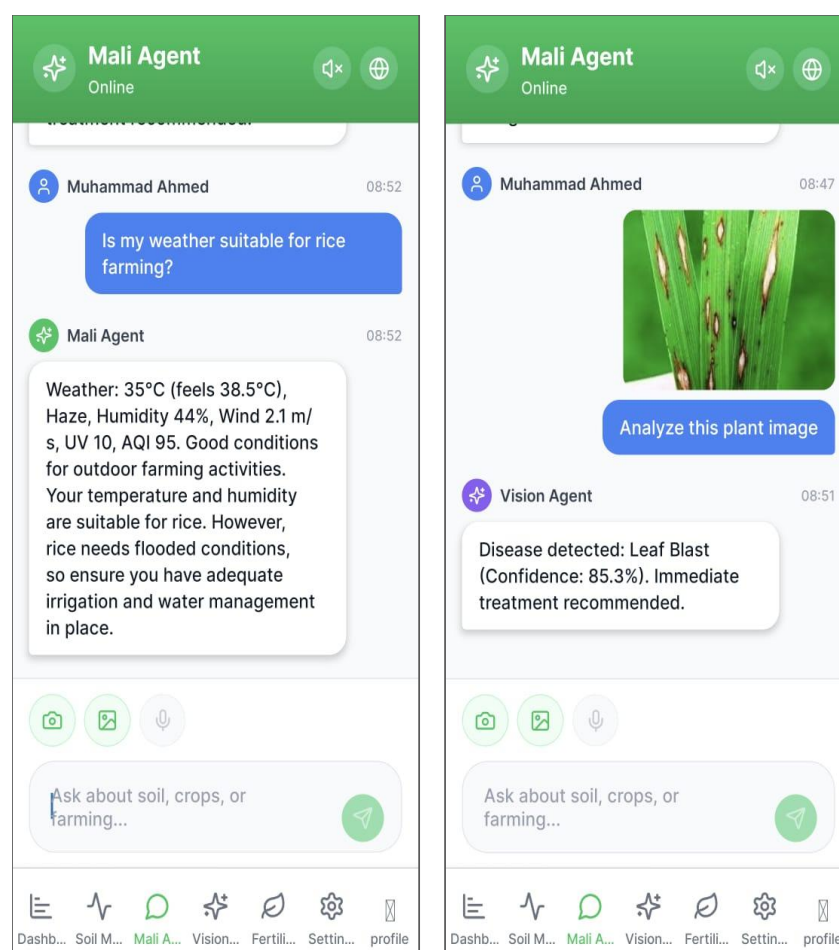
Figure 23. Vision agent: diffusion model + RiceNet confusion matrix.

5.4. Supervisor Agent

The supervisor agent serves as an integration layer that establishes a connection between the individual agents, ensuring that their outputs are not isolated, but rather combined into a unified response. The supervisor agent uses the open-source AI model LLM to generate intelligent and context-aware responses based on user queries, thereby acting as a connector between the system's agents and the end user. By doing so, it not only retrieves relevant insights from the soil, climate, and vision agents, but it also synthesizes these insights into meaningful recommendations that can assist farmers in making informed decisions. Table 9 illustrates the database output generated through this process, where the soil and climate agent values are combined and accompanied by suggested actions (if required). Meanwhile, Figure 24 presents the multi-agent interface running on a mobile application, showcasing how the supervisor agent integrates the soil agent, weather agent, and vision agent into a single interactive system that delivers real-time, user-friendly results.

Table 9. Agent database showing soil and weather actions for the agriculture fields of Karachi and Thatta.

Timestamp	Climate Action	Soil Action	Climate Values	Soil Values	Climate Priority	Soil Priority
23 May 2025 20:52	Recommend cooling	Urgent nitrogen fertilizer req	temp: 36, humd: 84	nit: 304, phos: 159, pot: 461	high	critical
23 May 2025 20:51	Recommend cooling	Reduce nitrogen application	temp: 36, humd: 80	nit: 706, phos: 168, pot: 97	high	medium
23 May 2025 20:50	Recommend decreasing	Reduce nitrogen application	temp: 25.9, humd: 82	nit: 990, phos: 63, pot: 392	high	medium
23 May 2025 20:49	Recommend cooling	Reduce phosphorus application	temp: 35.8, humd: 76	nit: 666, phos: 165, pot: 133	high	medium
23 May 2025 20:48	Recommend cooling	Reduce nitrogen application	temp: 35.5, humd: 72	nit: 790, phos: 253, pot: 145	high	medium

**Figure 24.** Supervisor agent: weather and soil conditions (left) and vision agent (right).

6. Discussion and Future Work

These results confirm that the agentic AI framework is efficient in supporting autonomous agricultural monitoring by combining sensing, predictions, and user interaction. The system being modular in character provides an opportunity to extend it by adding

more sophisticated agents able to complete complex farming tasks, such as early stress detection, automated irrigation control, or yield forecasting. Moreover, multi-modal inputs such as soil nutrients, environmental properties, and images of rice crop plants show great promise for increasing the accuracy of decision-making when treated by specific agents. Nevertheless, some challenges remain. The current system relies on a small number of sensors; its predictive performance could be further enhanced with larger amounts of data from wider geographical areas and crop varieties. Apart from that, various environmental factors may affect the sensor performances and prediction models, such as extreme weather conditions, shifting soil conditions, and seasonal cycles. Future work could focus on the following directions:

- The integration of additional sensors such as multispectral cameras, soil pH and EC sensors, wind-speed detectors, and real-time rainfall gauges for richer environmental understanding.
- The implementation of ensemble learning and state-of-the-art models to enhance the predictions for better decision-making.
- Adaptive on-device learning, enabling agents to fine-tune models using local field data without requiring cloud retraining.
- Improved user interaction, such as adding multilingual support for low-literate farmers and users.
- Scalability and generalization, including testing across multiple crops, regions, and soil types to make the framework robust for global deployment.

7. Conclusions

In this work, we propose an agent-based AI framework for the autonomous monitoring of agricultural land by fusing data from a variety of soil and environmental sensors with AI-driven analytics. The agents fetch the parameters of temperature, humidity, nitrogen, phosphorus, potassium, and rice crop plant imagery, then process these inputs into actionable insights for farmers and landowners. This framework demonstrates how agent-based systems can reduce manual labor, improve situational awareness, and foster data-driven farming practices. Our results prove that it is possible for agentic AI to efficiently streamline farming operations and enable the early detection of diseases, hence contributing to improved crop productivity. While this is promising, the addition of more diverse sensors, the enabling of autonomous control mechanisms, and the deployment of the framework at larger scales could lead to further improvement in its robustness. In general, the present work contributes to the growth of smart agriculture through a flexible and expandable agent-based solution that may evolve into a completely autonomous farming assistant in the future.

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(accessed on 1 November 2025). The sensor dataset, which includes soil and weather parameters, is private and can be made available by the corresponding author upon reasonable request.

Conflicts of Interest: All authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

1D-CNN	One-Dimensional Convolutional Neural Network
AI	Artificial Intelligence
API	Application Programming Interface
ChatGPT	Chat-Generative Pre-Trained Transformer
CNN	Convolutional Neural Network
EU	European Union
FC	Fully Connected
GHG	Greenhouse Gases
GRU	Gated Recurrent Unit
IoT	Internet of Things
IRRI	International Rice Research Institute
LLM	Large Language Model
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
NLP	Natural Language Processing
NPK	Nitrogen, Phosphorus, and Potassium
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
SDG	Sustainable Development Goal
SVM	Support Vector Machine
UN	United Nations
VAE	Variational Autoencoder
ViT	Vision Transformer

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